

FASTAPI FOR AI ENGINEERING

Turn Python into a real API for an AI application

— *one lab at a time.*

13

Hands-on labs

+1

Capstone project

0

Prerequisites needed

YOU'LL BUILD



Validated chat
endpoints



File uploads



Streaming &
real-time chat



Security &
rate limits



Background
tasks



Synthesis Project
+
Capstone

Read → Run → Build · No prerequisites

WHAT IS FASTAPI?

A Python web framework for building real APIs

API, in one line: an interface on your server — your app exposes URL endpoints, and other programs call them over HTTP to request data and get responses.

FastAPI is the framework that handles the request/response plumbing for those APIs — so you focus on your app's logic.

WHERE IT'S USED

One of the most popular choices for AI backends — chat, uploads, streaming, real-time features, auth.

HOW A REQUEST FLOWS



Client sends an HTTP request

GET, POST, etc. to a URL endpoint



FastAPI routes it

to the right Python function



Your function runs

and returns a response



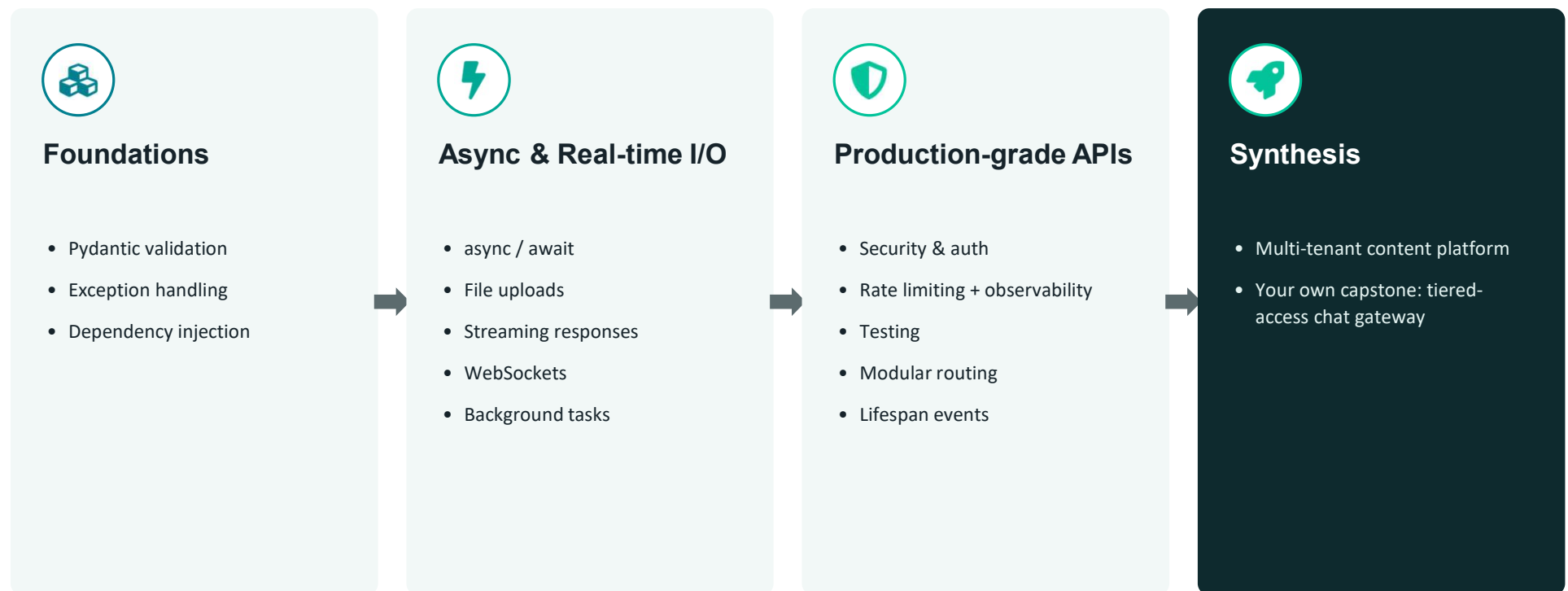
Docs, for free

interactive API docs at /docs

Every lab in this series builds exactly those capabilities on top of FastAPI.

THE ROADMAP

Same foundation, one new capability at a time



LAB STRUCTURE

Anatomy of one lab



Article

Explains the concept and what you'll build



Notebook

Run cell by cell — the first cell installs everything



Script

The same code as a .py file, for running as a real server



Assignment

A simple, step-by-step coding task you implement yourself



THE LOOP

Read

Run

Build

Next lab

RUNNING THE LABS

How to run the labs

See the reference file — it covers everything, end to end.



DEFAULT PATH

The Notebook

No server needed. Run cells top to bottom straight in your browser.

- Google Colab
- Jupyter
- VS Code



WHEN YOU WANT IT

The Real Server

The same code as a .py file, run with Uvicorn as a live server.

- `uvicorn main:app --reload`
- Test via the /docs page

GETTING STARTED

Labs aren't a rigid ladder

Each lab states what it assumes — several can be done in a different order.



Skim the reference file

Get oriented on how the labs run



Do Lab 1

Pydantic validation — your foundation



Continue with the lighter early labs

Exception handling, dependency injection — before the heavier async & file-upload lab that follows



Finish with the platform, then your capstone

Synthesis project, then design your own capstone